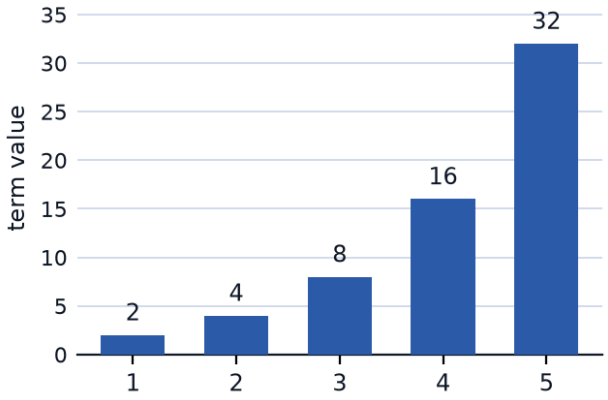
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Reading a sequence from its terms

The bar chart shows the first five terms of a sequence.

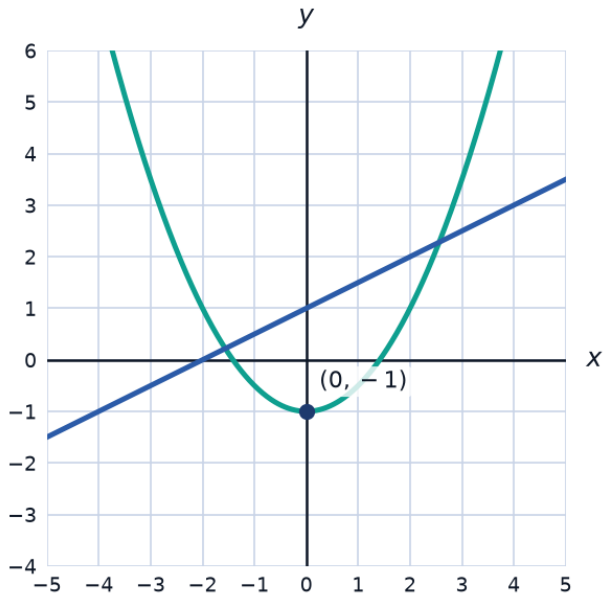


a Determine whether the sequence is arithmetic or geometric, and find the common ratio or difference.

b Find the 8th term.

2 [Foundational] • Target: Arithmetic sequences: finding terms

Find the 12th term of the arithmetic sequence with $a = 5$, $d = 3$.



3 [Foundational] • Target: Arithmetic sequences: finding terms

Find the 20th term of the arithmetic sequence with $a = -4$, $d = 2$.

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4 [Foundational] • Target: Arithmetic sequences: finding terms

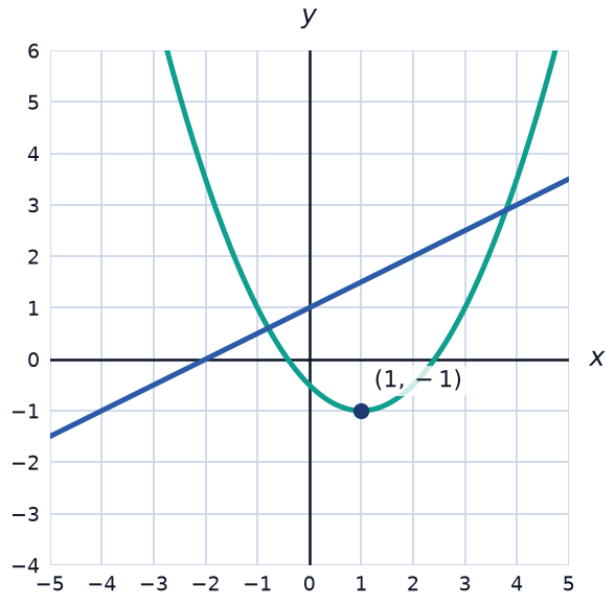
Find the 15th term of the arithmetic sequence with $a = 100$, $d = -6$.

5 [Foundational] • Target: Arithmetic sequences: finding terms

Write the first four terms of the arithmetic sequence with $a = 7$, $d = -2$.

6 [Foundational] • Target: Arithmetic sequences: finding a and d

The 4th term of an arithmetic sequence is 17 and the 9th term is 42. Find the first term a and common difference d.



7 [Foundational] • Target: Arithmetic sequences: finding a and d

An arithmetic sequence has $a_1 = 10$ and $a_6 = 30$. Find the common difference d.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Arithmetic sequences: finding a and d

An arithmetic sequence has $a_3 = 11$ and $a_7 = 27$. Find the common difference d.

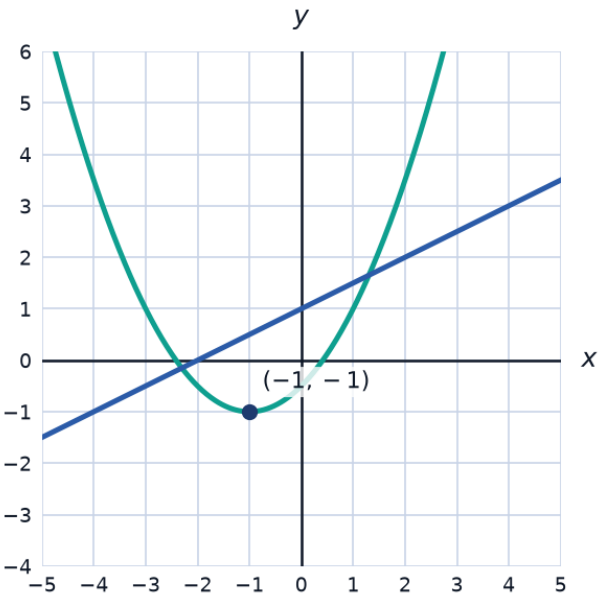
9 [Intermediate] • Target: Arithmetic series: the sum formula

Find the sum of the first 20 terms of the arithmetic sequence with $a = 3$, $d = 4$, using $S_n = \frac{n}{2}(2a + (n-1)d)$.

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10 [Intermediate] • Target: Arithmetic series: the sum formula

Find the sum of the first 15 terms of the arithmetic sequence with $a = 50$, $d = -3$.



11 [Intermediate] • Target: Arithmetic series: the sum formula

Find the sum of the arithmetic series $2 + 5 + 8 + \cdots + 50$.

12 [Intermediate] • Target: Arithmetic series: the sum formula

Find the sum of the first 10 positive multiples of 6.

13 [Intermediate] • Target: Geometric sequences: finding terms

Find the 6th term of the geometric sequence with $a = 3$, $r = 2$.

14 [Intermediate] • Target: Geometric sequences: finding terms

Find the 5th term of the geometric sequence with $a = 100$, $r = 0.5$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

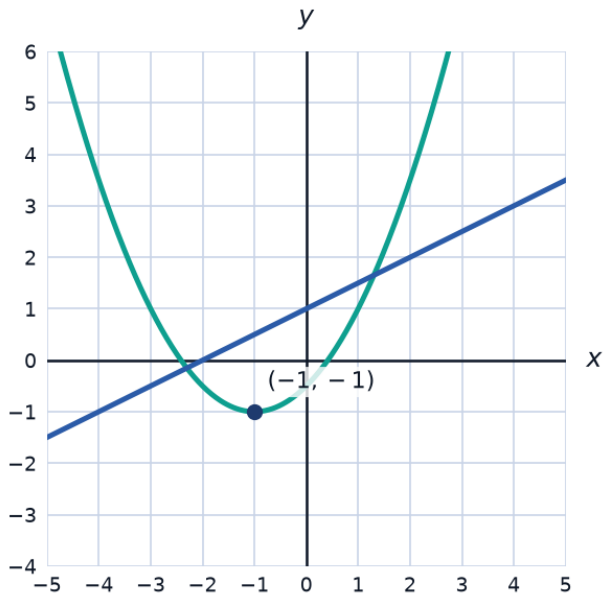
15 [Challenging] • Target: Geometric sequences: finding terms • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Find the 4th term of the geometric sequence with $a = 2$, $r = -3$.

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16 [Challenging] • Target: Geometric sequences: finding terms • Cross-Lesson: 6.01 Limits & Derivatives

Write the first four terms of the geometric sequence with $a = 5$, $r = 3$.



17 [Challenging] • Target: Geometric sequences: finding a and r • Cross-Lesson: 7.01 Integration & Applications

The 2nd term of a geometric sequence is 6 and the 5th term is 162. Find r and a.

18 [Challenging] • Target: Geometric sequences: finding a and r • Cross-Lesson: 8.01 Exponentials & Logarithms

A geometric sequence has $a_1 = 4$ and $a_4 = 108$. Find r.

19 [Challenging] • Target: Geometric sequences: finding a and r • Cross-Lesson: 9.01 Statistics & Probability

A geometric sequence has 3rd term 12 and 6th term 96. Find a and r.

20 [Challenging] • Target: Geometric series: the finite sum • Cross-Lesson: 1.01 Algebra & Equations

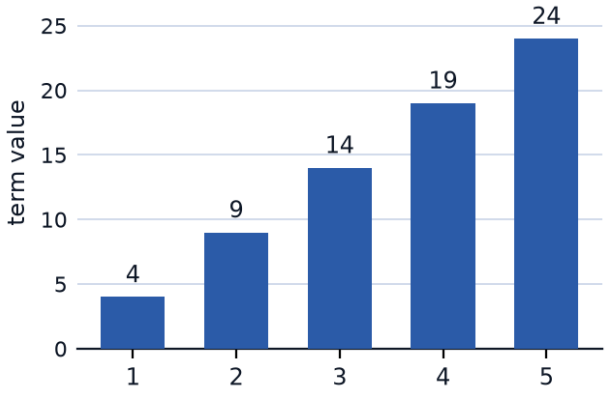
Find the sum of the first 6 terms of the geometric series with $a = 3$, $r = 2$, using $S_n = \frac{a(r^n - 1)}{r - 1}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: MCQ — Reading a sequence from its terms

The bar chart shows the first five terms of a sequence. Find the common difference.



a. 6

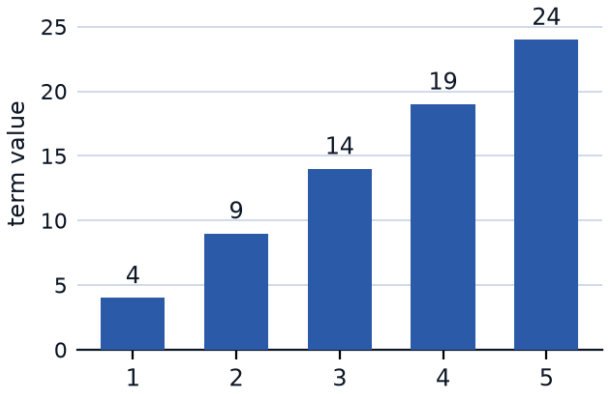
b. 5

c. 4

d. 9

22 [Foundational] • Target: MCQ — Reading a sequence from its terms

Using the same bar chart, predict the 8th term.



a. 29

b. 34

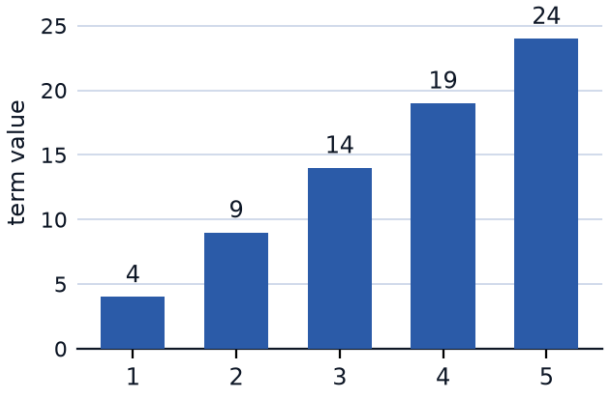
c. 39

d. 44

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23 [Foundational] • Target: MCQ — Reading a sequence from its terms

Using the same bar chart, find the sum of the first five terms shown.



a. 70

b. 66

c. 24

d. 75

24 [Foundational] • Target: MCQ — Arithmetic sequences: finding terms

Find the 14th term of the arithmetic sequence with $a = 6$, $d = 4$.

a. 62

b. 54

c. 50

d. 58

25 [Foundational] • Target: MCQ — Arithmetic sequences: finding terms

Find the 25th term of the arithmetic sequence with $a = 2$, $d = 7$.

a. 177

b. 170

c. 175

d. 163

26 [Foundational] • Target: MCQ — Arithmetic sequences: finding a and d

The 3rd term of an arithmetic sequence is 15 and the 8th term is 45. Find the first term a and common difference d.

a. $a = 3, d = 5$

b. $a = 3, d = 6$

c. $a = 6, d = 3$

d. $a = 9, d = 6$

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27 [Foundational] • Target: MCQ — Arithmetic series: the sum formula

Find the sum of the first 30 terms of the arithmetic sequence with $a = 1$, $d = 2$ (the odd numbers).

a. 960

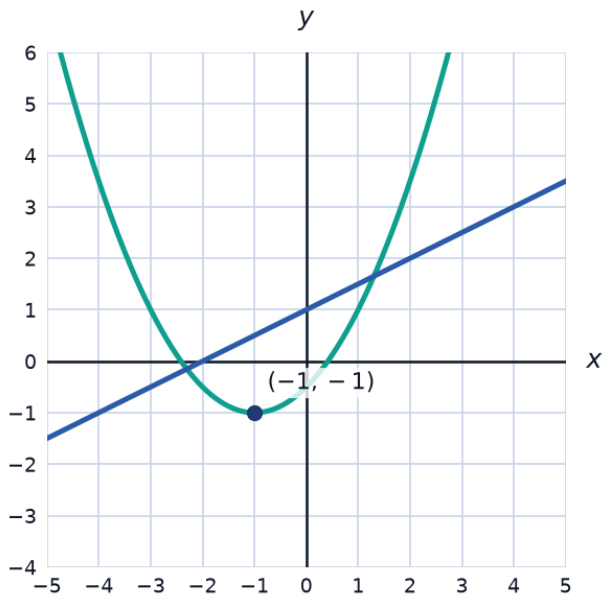
b. 930

c. 870

d. 900

28 [Foundational] • Target: MCQ — Geometric sequences: finding terms

Write the first four terms of the geometric sequence with $a = 4$, $r = 2$.



a. $4, 6, 8, 10$

b. $4, 16, 32, 64$

c. $4, 8, 12, 16$

d. $4, 8, 16, 32$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Geometric sequences: finding a and r

A geometric sequence has 2nd term 4 and 5th term -108. Find r.

a. $r = 3$

b. $r = -27$

c. $r = -3$

d. $r = -9$

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30 [Intermediate] • Target: MCQ — Geometric series: the finite sum

Find the sum of the first 8 terms of the geometric series with $a = 1$, $r = 2$.

a. 255

b. 128

c. 254

d. 256

31 [Intermediate] • Target: MCQ — Infinite geometric series

Express the repeating decimal $0.777\ldots$ as a fraction using an infinite geometric series with $a = 0.7$, $r = 0.1$.

a. $\frac{7}{90}$

b. $\frac{77}{100}$

c. $\frac{7}{9}$

d. $\frac{7}{10}$

32 [Intermediate] • Target: MCQ — Sigma notation

Use $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ to evaluate $\sum_{k=1}^{15} k$.

a. 120

b. 110

c. 105

d. 136

33 [Intermediate] • Target: MCQ — Recursive sequences

A sequence satisfies $a_1 = 2$ and $a_{n+1} = 2a_n + 1$. Find a_4.

a. 11

b. 23

c. 5

d. 47

34 [Intermediate] • Target: MCQ — Identifying arithmetic and geometric sequences

State whether $5, 10, 20, 40, \ldots$ is arithmetic or geometric, and give the common difference or ratio.

a. Geometric, $r = 2$

b. Geometric, $r = 5$

c. Arithmetic, $d = 2$

d. Arithmetic, $d = 5$

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35 [Intermediate] • Target: MCQ — Arithmetic series: the sum formula

Find the 16th term of the arithmetic sequence with $a = 20$, $d = -7$.

a. -82

b. -92

c. -85

d. -78

36 [Intermediate] • Target: MCQ — Geometric sequences: finding terms

Find the 13th term of the arithmetic sequence with $a = -16$, $d = 9$.

a. 95

b. 101

c. 83

d. 92

37 [Intermediate] • Target: MCQ — Geometric sequences: finding a and r

Find the 8th term of the arithmetic sequence with $a = 6$, $d = 2$.

a. 23

b. 22

c. 18

d. 20

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Geometric series: the finite sum • Cross-Lesson: 8.01 Exponentials & Logarithms

Find the 5th term of the arithmetic sequence with $a = -2$, $d = 9$.

a. 25

b. 43

c. 37

d. 34

39 [Challenging] • Target: MCQ — Infinite geometric series • Cross-Lesson: 9.01 Statistics & Probability

Find the 6th term of the geometric sequence with $a = -9$, $r = -2$.

a. 291

b. -288

c. 285

d. 288

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40 [Challenging] • Target: MCQ — Sigma notation • Cross-Lesson: 1.01 Algebra & Equations

Find the 4th term of the geometric sequence with $a = 1, r = -3$.

a. -27

b. -30

c. 27

d. -24

41 [Challenging] • Target: MCQ — Recursive sequences • Cross-Lesson: 2.01 Functions & Transformations

Find the 7th term of the geometric sequence with $a = -7, r = -2$.

a. 448

b. -445

c. -448

d. -451

42 [Challenging] • Target: MCQ — Identifying arithmetic and geometric sequences • Cross-Lesson: 4.01 Trigonometry

Find the sum of the infinite geometric series with $a = 16, r = \frac{1}{4}$.

a. $\frac{64}{3}$

b. $-\frac{64}{3}$

c. $\frac{58}{3}$

d. $\frac{70}{3}$

43 [Challenging] • Target: MCQ — Reading a sequence from its terms • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Evaluate $\sum_{k=1}^{6} (2k + 2)$.

a. -54

b. 54

c. 50

d. 58

44 [Challenging] • Target: MCQ — Word problems: sequences in context • Cross-Lesson: 6.01 Limits & Derivatives

A sequence satisfies $a_1 = 2$ and $a_{n+1} = a_n + 4$. Find a_4.

a. 10

b. 14

c. 17

d. 11

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45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 7.01 Integration & Applications

Find the sum of the first 7 terms of the geometric series with $a = 2$, $r = 2$, using $S_n = \frac{a(r^n - 1)}{r - 1}$.

a. 256

b. 508

c. 254

d. 127

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